Study Documents for "Towards Developer Engagement with Requirements during Code Review"

CONTENTS

I Study Materials 1
- I-A Consent Letter . . . 1
- I-B Subject Information Form . . . 2
- I-C Laboratory In-Person: Code Review . . . 3
- I-D Email Recruitment . . . 4

II Merge Request Information and Analysis 5

I. STUDY MATERIALS

A. Consent Letter

Redacted due to double anonymous submission.

B. Subject Information Form

Once we've obtained informed consent, we will ask participants to complete the following information sheet on paper. It will be stored with the consent forms.

PARTICIPANT DATA SHEET

Thank you for agreeing to participate in our study, entitled "Investigating Agile Developer Actions".
The purpose of this form is so that we can describe general descriptive/aggregate statistics about our study cohort (e.g., traditionally college aged students, breakdown by class year, percentage with prior programing internship). Raw data from this form will NOT be shared and will be kept confidential.

What is your name as it appears on your consent form?

What is your age as of September 1, 2025?

What is your current/expected graduation date (e.g., 2027, 2027J)?

How many credits have you completed as of September 1, 2025?

Have you previously had a technical or software development internship? If so, where and what was your roll?

Have you previously used git, gitHub, or gitLab? If so, in what context?

What programming languages do have you experience with? Please list each language and rate your proficiency on the following scale: *1. Beginner/Novice, 2. Advanced Beginner, 3. Intermediate/Competent, 4. Advanced/Proficient, 5. Expert.*

C. Laboratory In-Person: Code Review

In this part, we ask participants to review and comment on a pull/merge request. This process involves them reviewing the code, adding a few comments and deciding whether it should be approved (merged into the master codebase), revised (sent back to the developer for further improvement), or denied (rejected entirely).
Participants will be asked to repeat this activity multiple times throughout the semester. To mitigate learning and carryover effects, we vary which codebase they review in this investigation and whether the code is buggy or not.
The main experimental condition under investigation is the presence of requirements, with the following three options:

- *No requirements present.*
- *The entire requirements documents present, unsorted.*
- *List of requirements presented (unordered, ordered, or intentionally mis-ordered).*

Intro: Thanks for coming in today to complete a code review. We will be working with the "insert repository name" codebase. I understand that you should be relatively familiar with this code because you reviewed it during CSC400. Do you have any unanswered questions from the class component?
We identified a "bug fix or feature request" that we would like you to evaluate and perform a code review on.
(*If eye tracking available: 'We are going to try using an eye tracker in this session. We will ask you to "insert task, likely follow a ball on the screen" to calibrate the eye tracker. After the machine is calibrated, we ask you to sit relatively still and not move the position of your chair, so please make sure you are comfortable now.'*)
We will start the screen recording now, so that we can record how you interact with the git interface and the code. Do we have your consent to record the audio for this session to help us take notes? *If consent received, start audio recording and reaffirm consent once recording.*

Review: Let's start by reviewing the merge request and making sure you understand it. Please take a few moments to review it. The code is available "insert tool" and the "insert requirements scenarios". We're here to help you and answer any of your questions. It would be helpful for us if you can describe any thoughts that you have as you review the pull request. Questions that could be asked while the participants work on the code review:

- What are you thinking about?
- What comments would you like to leave for the developers?
- Are there any parts that you think require further investigation?
- Would you prefer a different implementation?

Debrief: We will now stop the screen (and audio) recording. Now that you have completed the code review, we ask you to complete a short questionnaire (on Qualtrics) about this code review.

- Please enter your chosen pseudonym.
- Rate your level of confidence in your review: Very confident, Somewhat confident, Neither confident nor unconfident, Somewhat unconfident, and Not at all confident.
- To what extent did you exhaustively test the code affected by the pull request?
- Does the update satisfy the requirements? Why or Why not?
- Any additional comments about your participation?

Thank you for your time.

D. Email Recruitment

Redacted due to double anonymous submission.

II. Merge Request Information and Analysis

In this section, we provide additional details about the merge requests. Tbl. I lists the details for each time category in RQ1. Each subject (A through J) reviewed one of five BloomingLeaf merge requests without access to the tool's requirements documents for Period 1. In Period 2, subjects reviewed the same request for Strawbeberya with access to the project requirements. Tbl. II lists the metrics for each merge request used in this study.

TABLE I: Categories for Time Spent in Merge Reviews

Code	Code encapsulates time spent looking at code either in the 'Changes' tab of GitLab, externally on VSCode, or in a split screen with these two windows open.
Testing	Testing describes time during which the user ran the program and was actively manipulating it.
MR Overview	MR Overviewrefers to the landing page of the merge request, in which the title, description, and user information is visible.
Issue	Issue indicates time spent looking at the associated issue, which was linked from the 'Overview' page.
Commits	Commits refers to the 'Commits' tab of GitLab which lists the sequence of titles of the commits in the merge request.
Writing Review	Writing review indicates time during which the user is typing their comment.
Other	Other consists of activity in which the Subject's focus was unable to be determined, such as while pages were loading or while the Subject was idle on a split screen with multiple windows open.
Requirements	Requirements refers to time spent looking at the requirements list.

TABLE II: Merge Request Review Data

Period 1 - BloomingLeaf					
ID	Files	LOC	Title	Description	Subjects
MR1	1	+7 -4	Updated Softgoal Symbol	Changed the shape of softgoal to the shape of goal and added vertical lines while maintaining scalability. Closes #3	A, E, G
MR2	3	+41 -3	Color Relationship Links By Type	Colors AND (green), OR (yellow), HELP (blue), and HURT (red) links differently to more clearly distinguish them. Updated to be cross-compatible with Issue #8 (updating link names).	B, C, J
MR3	3	+109 -1	Allow user to focus on dependencies of one intention	See Dependencies' toggle created to see the dependencies associated with an intention. The dependencies are coloured grey when the toggle is turned on.	D
MR4	4	+167 -80	Toggle between beginner and advanced modes	This creates a toggle at the user interface that controls whether in beginner mode or advanced mode. The default is in advanced mode. Modified in index.html and style.css, and added beginnerObjects.js.	F, H
MR5	2	+239 -117	Added labels to intentions	Added small textual labels to each intention specifying their type (goal, task, softgoal, resource). Closes #1	I
Period 2 - Strawbeberya					
	Files	LOC	Title	Description	Subjects
	1	+27 -7	Add outline around selected boxSelection box	When the user clicks on a fruit, adds an outline around that box to show which fruits are part of the current guess.	ALL